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Cryobiopsy vs Forceps for Bronchoscopic Lung Biopsy
The FROSTBITE-2 Randomized Clinical Trial

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IMPORTANCE Bronchoscopic biopsy is conventionally performed with forceps, which can result in small specimen sizes and poor specimen quality due to crush artifact. Cryoprobe use localizes freezing at the probe tip, enabling retrieval of larger, more intact biopsy specimens.

OBJECTIVE To evaluate the diagnostic yield of a 1.1-mm cryoprobe for transbronchial biopsy.

DESIGN, SETTING, AND PARTICIPANTS This open-label, outcome assessor–masked, multicenter randomized clinical trial included 500 patients aged 18 years or older scheduled to undergo transbronchial biopsy for lung nodules or masses, lung transplant, or diffuse parenchymal lung disease. The trial was conducted in 9 US medical centers and enrolled patients between February 27, 2023, and September 11, 2024. The date of last follow-up was October 12, 2024.

INTERVENTION Patients were randomized 1:1 to transbronchial biopsy using a 1.1-mm cryoprobe (n = 250) or 2.0-mm forceps (n = 250).

MAIN OUTCOMES AND MEASURES The primary outcome was diagnostic yield, defined as the percentage of patients for whom the transbronchial biopsy sample led to a specific diagnosis based on histologic examination. Of the 8 prespecified secondary analyses, key secondary analyses were the diagnostic yield for each of the 3 conditions (lung nodules or masses, lung transplant, and diffuse parenchymal lung disease) and complication rates.

RESULTS Of 774 patients assessed for eligibility, 609 provided consent, 500 were randomized, and 490 were included in the primary analysis; the mean age was 62.6 years (SD, 12.7 years) and 252 of 500 (50.4%) were male. The primary outcome of diagnostic yield was significantly higher in patients randomized to transbronchial biopsy with cryoprobes vs forceps (217 of 245 [88.6%] vs 193 of 245 [78.8%]; absolute difference, 9.8%; 95% CI, 3.3%-16.3%; *P* = .003). For the key secondary analyses, compared with that of forceps, the diagnostic yield of cryoprobes was significantly higher among patients with pulmonary nodules or masses (79 of 95 [83.2%] vs 68 of 97 [70.1%]; absolute difference, 13.1%; 95% CI, 1.0%-24.6%; *P* = .04) and lung transplant (120 of 125 [96.0%] vs 110 of 124 [88.7%]; absolute difference, 7.3%; 95% CI, 0.6%-14.4%; *P* = .03) but did not differ significantly in diffuse parenchymal lung disease (18 of 25 [72.0%] vs 15 of 24 [62.5%]; absolute difference, 9.5%; 95% CI, −16.0% to 33.6%; *P* = .55). For the secondary safety analysis, there were 4 pneumothoraces requiring chest tube placement in the forceps group (1.6%) vs none in the cryoprobe group; no patients experienced significant bleeding or respiratory failure events.

CONCLUSIONS AND RELEVANCE Transbronchial lung biopsy performed with a 1.1-mm cryoprobe had a significantly higher diagnostic yield compared with 2.0-mm forceps in a group of patients with lung nodules or masses, lung transplant, and diffuse parenchymal lung disease.

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Visual Abstract
Research Summary
Supplemental content

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Bronchoscopy is a common procedure, with more than 500 000 bronchoscopies and an estimated 190 000 transbronchial biopsies performed annually in the United States.[1,2] Despite recent technologic advancements, including robotic-assisted bronchoscopy and intraprocedural 3-dimensional imaging demonstrating improvements in the ability to navigate the lung periphery with high precision,[3,4] diagnostic yields of transbronchial biopsy remain 55% to 70%, depending on the indication for bronchoscopy.[5-9]

The standard method to obtain tissue during transbronchial biopsy is with use of forceps, which are passed through the working channel of the bronchoscope to the target lung tissue. After the forceps are opened and advanced, their jaws grasp lung tissue. Then the jaws are closed to cut and retrieve tissue for analysis.[1] A more recent technique for transbronchial biopsy involves cryoprobes, which use localized freezing at the probe tip to obtain lung tissue samples through cold adhesion. Prior studies using a 1.9-mm cryoprobe for transbronchial biopsy in patients with suspected usual interstitial pneumonia have shown higher diagnostic yields than with forceps.[2,3,7-9] However, initial reports of cryobiopsy use with larger cryoprobes (1.9 and 2.4 mm) led to safety concerns due to increased rates of bleeding and pneumothoraces after transbronchial biopsy[10] because the cryoprobe and bronchoscope need to be removed en bloc.[10-12]

A smaller 1.1-mm flexible cryoprobe has been designed for lung tissue retrieval through the working channel of the bronchoscope, as with forceps, allowing the bronchoscope to remain in place to observe and manage any potential complications. This device was approved by the US Food and Drug Administration in 2020, and prior studies have shown its safety profile to be noninferior to forceps.[11,13,14] The FROSTBITE-1 trial, which was a prospective, single-group study of 50 consecutive patients referred for transbronchial biopsy, demonstrated that use of the 1.1-mm cryoprobe had complication rates similar to those of standard forceps.[13] However, to our knowledge no randomized clinical trials have been published comparing the diagnostic yield of a 1.1-mm flexible cryoprobe with forceps, so differences in diagnostic yield have not been established.

The FROSTBITE-2 trial was designed to test the hypothesis that the diagnostic yield would be higher with use of a 1.1-mm flexible single-use cryoprobe vs standard forceps biopsy, without differences in safety outcomes.

Key Points

Question Is the diagnostic yield of transbronchial lung biopsy higher with use of a 1.1-mm cryoprobe compared with 2.0-mm forceps?

Findings In this randomized clinical trial of 500 patients undergoing transbronchial biopsy for pulmonary nodules or masses, evaluation after lung transplant, or diffuse parenchymal lung disease, the diagnostic yield was 88.6% with cryobiopsy vs 78.8% with forceps (absolute difference, 9.8%).

Meaning The diagnostic yield of transbronchial lung biopsy was significantly higher with use of a 1.1-mm cryoprobe vs 2.0-mm forceps in a group of patients with pulmonary nodules or masses, lung transplant, and diffuse parenchymal lung disease.

Methods

Trial Design and Oversight

This randomized clinical trial was conducted in accordance with Regulations for the Protection of Human Subjects and the International Conference on Harmonization Good Clinical Practice guidelines and received institutional review board approval at the coordinating center, Johns Hopkins University, and at each of the trial sites. All participants provided written informed consent. Trial oversight was provided by an independent data and safety monitoring board. This report follows the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline[15] and was registered at ClinicalTrials.gov (NCT05751278) on February 20, 2023.

Trial Population

Eligible patients were aged 18 years or older and scheduled to undergo transbronchial biopsy according to standard of care for lung nodule or mass, lung transplant evaluation, and diffuse parenchymal lung disease. Exclusions were consistent with standard contraindications to transbronchial biopsy, including presence of a bleeding disorder, platelet count less than $50 \times 10^3/\mu L$ at enrollment, use of systemic anticoagulation or antiplatelet therapy without the ability to hold therapy for the recommended time before the procedure (aspirin monotherapy was acceptable), and patients with do-not-resuscitate or do-not-intubate status. Race and ethnicity were self-reported in closed categories.

Trial Design, Treatments, and Assessments

FROSTBITE-2 was an investigator-initiated, open-label, masked (outcome assessors) randomized clinical trial conducted at 9 US medical centers that performed at least 100 transbronchial biopsies per year and had affiliated institutional centers for lung cancer, lung transplant, and interstitial lung disease. Details of the trial protocol, including the statistical analysis plan, have previously been published[16]; the trial protocol is shown in Supplement 1.

Enrollment, Randomization, and Masking

Consecutive patients who were referred for transbronchial biopsy were screened by the local site investigator. After informed written consent, participants were randomly assigned in a 1:1 ratio to transbronchial biopsy using 2.0-mm forceps or a 1.1-mm cryoprobe. Randomization was stratified by indication (lung transplant, lung nodule or mass, and diffuse parenchymal lung disease) with the use of a web-based randomization tool.[17-19] Patients with pulmonary nodules or masses were randomized intraprocedurally once the decision was made to perform a transbronchial biopsy. Patients who were enrolled but achieved a diagnosis by another method (eg, transbronchial needle aspiration) during the bronchoscopy were considered screen failures and not randomized. Given the nature of intervention, treating bronchoscopists and research personnel were not masked to the trial group assignments after randomization; however, outcome assessors (histopathology) were masked to trial group assignments.

Trial Intervention

All bronchoscopy procedures were performed consistent with standard of care according to local institutional preferences, aside from randomization to the technique used to obtain the biopsy specimen (cryoprobe or forceps). All investigators for each site must have completed 5 procedures with the 1.1-mm cryoprobe before the study. Any participating investigator without this experience completed study-specific training with the 1.1-mm cryoprobe and at least 1 directly observed case by the study principal investigator before enrolling any patients. Patients underwent the procedure under deep sedation or general anesthesia with an artificial airway (endotracheal tube, laryngeal mask airway, or tracheostomy) in place according to standard of care. Use of a rigid bronchoscope was left to the discretion of the physician performing the procedure. Sedative and analgesic medications were administered by an anesthesiologist or procedural sedation expert in accordance with standard of care.

Other bronchoscopic sampling procedures, including bronchoalveolar lavage, transbronchial brushing, transbronchial needle aspiration, and endobronchial ultrasonographic transbronchial needle aspiration, were performed at the discretion of the bronchoscopist. According to standard practice, localization techniques, including fluoroscopy, radial endobronchial ultrasonography, electromagnetic navigation, robotic-assisted bronchoscopy, and cone beam computed tomography, could be used in both groups during bronchoscopy to aid localization because they are compatible with both standard-capacity 2.0-mm forceps and the 1.1-mm cryoprobe.

For participants randomized to the cryoprobe group, a 1.1-mm cryoprobe (Erbe Elektromedizin GmbH) was used, and for those randomized to the forceps group, locally available standard-capacity forceps were used, compatible with a 2.0-mm bronchoscope working channel. Forceps with an outer diameter of 1.7 to 2.0 mm were permitted, with the brand and jaw type left to the discretion of the local bronchoscopist.

Before induction of anesthesia for patients randomized to the cryoprobe group, the function of the cryoprobe was tested by immersing its tip in sterile water or isotonic saline at approximately 20 °C and activating the freeze function for approximately 5 seconds. The cryoprobe was deemed functional if a clearly visible, homogenous ball of ice formed at the tip to indicate adequate freezing capacity, with no visible gas bubbles. If this outcome was not achieved, a new cryoprobe was tested and used.

After anesthesia induction, the bronchoscope was advanced through the airway to the bronchus where the transbronchial biopsy would be performed. The cryoprobe was then inserted through the bronchoscope and gently advanced to the target tissue and activated for an initial freeze time of 3 seconds, measured on the device timer. If inadequate tissue was obtained because of incomplete freeze, the freeze time was extended by 1 second on each consecutive attempt (to a maximum of 8 seconds). The transbronchial biopsy sample was then extracted through the working channel, with the bronchoscope remaining in the airway, and the specimen was immediately placed in a liquid medium.

For patients randomized to the forceps biopsy, the bronchoscope was advanced into the airway to the bronchus where the transbronchial biopsy would be performed. The forceps were then inserted through the working channel of the bronchoscope and advanced until the targeted area was reached. The forceps were gently advanced under fluoroscopic guidance (if used) until slight resistance was met. After withdrawing the forceps 1 to 2 cm, the bronchoscopist opened the forceps jaws, advanced the forceps slightly, and closed the jaws to obtain a small piece of lung tissue. The forceps were then removed through the working channel of the bronchoscope, and the specimen was immediately placed in a liquid medium.

The number of biopsies taken was standardized. Patients in the lung transplant group had 10 transbronchial biopsy specimens collected according to International Society for Heart and Lung Transplantation guidelines,[20] and individuals with lung nodule or mass or diffuse parenchymal lung disease had 5 specimens obtained.

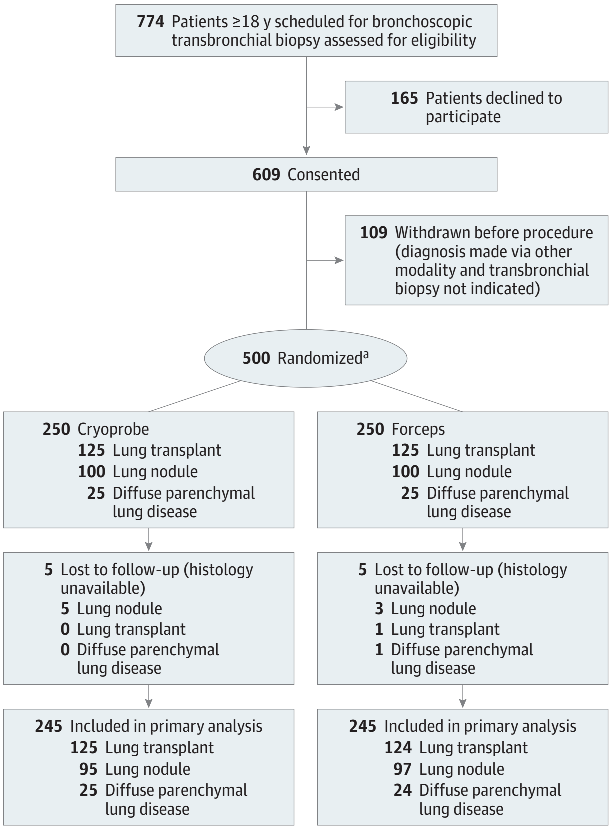
Collected specimens were sent to the local site's pathology laboratory for standard processing and clinical interpretation. All patients underwent immediate postprocedure chest radiography to assess for complications such as pneumothorax. At 30 days after the procedure, patients were contacted via telephone, medical records were reviewed to assess for adverse events, or both. All biopsies were initially locally processed with standard procedures and were interpreted by a local pathologist(s). All transbronchial biopsy specimens were digitized for centralized pathologic adjudication and then sent for centralized review to decrease heterogeneity in histopathologic interpretation between study sites and individual local pathologists. A central pathology core at Johns Hopkins University, masked to the transbronchial biopsy technique and local site pathology interpretation, assessed the primary outcome of diagnostic yield, provided the final histopathologic diagnosis, and determined whether the biopsy was diagnostic. The central pathology core provided pathologic interpretation and quantification of each specimen, assessing diagnostic yield and secondary histologic outcomes of the histologic accessibility index of the biopsy specimen (a 7-point Likert scale of specimen quality used in other studies of cryobiopsy specimens[11,21]), and quantified histologic parameters of total specimen area, total alveolated area, crush artifact proportion, and artifact-free proportion.

Study Outcomes

The primary outcome was the overall diagnostic yield based on transbronchial biopsy as assessed by the central pathology core, excluding samples obtained by other methods during the bronchoscopy (eg, endobronchial ultrasonographic transbronchial needle aspiration), and was defined as the percentage of patients for whom the transbronchial biopsy sample led to a specific diagnosis based on histologic examination.

Diagnostic criteria were predefined and depended on the indication for biopsy. In the lung nodule or mass group, strict

Figure. Flowchart of Patient Selection, Randomization, and Progression Through the Trial



[a]Stratified by indication.

diagnostic definitions as outlined by the American Thoracic Society were used.[22] A biopsy that resulted in a malignant or benign (eg, granuloma, infection) histopathologic diagnosis was considered diagnostic. Atypia or lung parenchyma without pathologic findings was considered nondiagnostic. For the diffuse parenchymal lung disease group, a biopsy result with a specific histopathologic pattern of inflammation (eg, usual interstitial pneumonia, organizing pneumonia, hypersensitivity pneumonitis) was considered diagnostic.[23] Nonspecific inflammation, inadequate tissue, or normal lung parenchyma was considered nondiagnostic. For the lung allograft group, a biopsy result that demonstrated any degree of acute cellular rejection from A0 (no rejection) to A4 (severe rejection) according to the criteria of the International Society for Heart and Lung Transplantation or another specific histopathologic diagnosis (eg, granuloma, infection, adenocarcinoma) was considered diagnostic.[20] Biopsies that were ungradable or inadequate for diagnosis of acute cellular rejection (Ax) were considered nondiagnostic.

There were 8 prespecified secondary analyses and outcomes: the diagnostic yield of each of the 3 conditions (lung nodules or masses, lung transplant, and diffuse parenchymal lung disease), histologic accessibility index of the biopsy specimen (a 7-point Likert scale of specimen quality used in prior studies of cryobiopsy specimens[11,21]), and measured histologic parameters of total specimen area, total alveolated area, crush artifact proportion, and artifact-free proportion. For assessment of safety outcomes, all adverse events and serious adverse events were assessed intraprocedurally, immediately after the procedure, and for up to 30 days after the procedure. Standardized scales according to the Common Terminology Criteria for Adverse Events (CTCAE; National Cancer Institute, version 5.0) were used to report adverse events related to bleeding, pneumothorax, and respiratory failure. A predefined composite safety end point was assessed and comprised bleeding (grade ≥3 modified CTCAE score), pneumothorax requiring chest drainage (grade ≥2 CTCAE score), postprocedural respiratory failure (defined as the need for noninvasive or mechanical ventilation requiring intensive care unit admission), and death within 30 days.

Statistical Analysis

Demographic and procedural features were summarized for each trial group, using median and quartiles for continuous variables and frequencies and percentages for categorical variables. The study was powered to detect differences in the overall diagnostic yield and in each of the subgroups, with 80% power and a significance level of .05. *P* values were 2-sided; analysis was performed with Stata version 13 (StataCorp) from January 16, 2026, to February 17, 2026. For lung nodules or masses, the diagnostic yield of forceps was assumed to be 55%, with a hypothesized increase of 20% with use of a cryoprobe,[7,8,24] with a 10% dropout rate, resulting in an estimated sample size of 200 patients. For lung transplant, the diagnostic yield of forceps was assumed to be 72%, with a hypothesized increase of 15% with the use of a cryoprobe,[25] with a 10% dropout rate, resulting in an estimated sample size of 250 patients. For diffuse parenchymal lung disease, the diagnostic yield of forceps was assumed to be 40%, with a hypothesized increase of 40% with the use of a cryoprobe,[26-31] with a 10% dropout rate, resulting in an estimated sample size of 50 patients. With the expectation of including 450 participants in the analysis after 10% dropout and using 70% as a conservative estimate for composite diagnostic yield of forceps, this study was powered to detect a minimum difference of 11.3% in overall diagnostic yield at a power level of 80%.

All participants with data available for the primary outcome were included and analyzed according to their allocation group. Fisher exact test was used to compare proportions and Wilcoxon rank sum tests were used to compare continuous variables. There was no multiplicity adjustment for the primary outcomes. A post hoc analysis for interaction (heterogeneity of effect) within the subgroups was performed. Participants without histology slides available for centralized review were not included in the primary analysis but were accounted for in a post hoc sensitivity analysis to explain the effect of missing data for the primary outcomes and secondary analysis of diagnostic yield. As part of the sensitivity analysis, missing data were imputed for best case (all diagnostic) and worst case (all nondiagnostic) in the respective groups, and differences between groups were assessed.

Results

Patient and Procedural Characteristics

Between February 27, 2023, and September 11, 2024, 774 patients were assessed for study eligibility at 9 US medical centers (**Figure**), with the date of last follow-up occurring on October 12, 2024. Among the 609 patients who consented, 109 did not undergo randomization because a diagnosis was made during bronchoscopy before transbronchial biopsy; of the remaining 500 patients who underwent transbronchial biopsy, 250 were randomized to the 1.1-mm cryoprobe (125 for lung transplant, 100 for lung nodules or masses, and 25 for diffuse parenchymal lung disease) and 250 patients were randomized to 2.0-mm forceps (125 for lung transplant, 100 for lung nodules or masses, and 25 for diffuse parenchymal lung disease). Ten of the 500 patients (2%) were considered lost to follow-up for the final analysis because of the absence of definitive histologic material for central pathologic review, which primarily occurred when participating centers could not locate or access archived pathology samples. As a result, these cases could not undergo standardized central review and were excluded from the final analytic cohort, 5 in the forceps group (1 lung transplant, 3 lung nodules or masses, and 1 diffuse parenchymal lung disease) and 5 in the cryoprobe group (5 lung nodules or masses), leaving 245 participants (98%) in each group with complete data for analysis of the primary outcome.

The demographic and procedural characteristics of patients are shown in **Table 1**. The mean age was 62.6 years (SD, 12.7 years), 252 of 500 participants (50.4%) were male, and 248 (49.6%) were female. Fifteen participants were Asian (3.0%), 71 were Black (14.2%), 387 were White (77.4%), and 27 were of unknown or unreported race (5.4%). Twenty-one participants were Hispanic (4.2%), 463 were non-Hispanic (92.6%), and 16 were of unknown or unreported ethnicity (3.2%). Most bronchoscopies were performed with either a laryngeal mask airway (278 of 500 [56%]) or endotracheal tube (220 of 500 [44%]). Only 0.6% of procedures (3 of 500) used a rigid bronchoscope, and 0.4% (2 of 500) used a prophylactic bronchial blocker.

Table 1. Patient and Procedural Characteristics

Characteristic	Cryoprobe (n = 250)	Forceps (n = 250)
Patient characteristics		
Median age (IQR), y	64 (57-71)	65 (58-71)
Sex, No. (%)		
Male	124 (49.6)	128 (51.2)
Female	126 (50.4)	122 (48.8)
Race, No. (%)[a]		
Asian	7 (2.8)	8 (3.2)
Black	26 (10.4)	45 (18)
White	201 (80.4)	186 (74.4)
Unknown or not reported	16 (6.4)	11 (4.4)
Ethnicity, No. (%)[a]		
Hispanic	14 (5.6)	7 (2.8)
Non-Hispanic	229 (91.6)	234 (93.6)
Unknown or not reported	7 (2.8)	9 (3.6)
Body mass index, median (IQR)[a]	25.8 (22.0-30.0)	26.7 (22.6-30.6)
Smoking status, No. (%)		
Currently	15 (6.0)	20 (8.0)
Formerly	128 (51.2)	124 (49.6)
Procedural characteristics, No. (%)		
Artificial airway		
Laryngeal mask airway	135 (54.0)	143 (57.2)
Endotracheal tube	114 (45.6)	106 (42.4)
Rigid bronchoscope	1 (0.4)	2 (0.8)
Bronchial blocker	1 (0.4)	1 (0.4)
Fluoroscopy	178 (71.2)	181 (72.4)

[a] Race, ethnicity, and body mass index are self-reported, closed categories. Body mass index is calculated as weight in kilograms divided by height in meters squared.

Primary Outcome

The primary outcome of diagnostic yield was 88.6% (217 of 245) in the cryoprobe group vs 78.8% (193 of 245) in the forceps group, for an absolute difference of 9.8% (95% CI, 3.3%-16.3%; *P* = .003) (**Table 2**).

Secondary Analyses

For the lung transplant group, the diagnostic yield was significantly higher with use of a cryoprobe compared with forceps (120 of 125 [96.0%] vs 110 of 124 [88.7%]; absolute difference, 7.3%; 95% CI, 0.6%-14.4%; *P* = .03). In the lung nodule or mass group, the diagnostic yield was also significantly higher with a cryoprobe vs forceps (79 of 95 [83.2%] vs 68 of 97 [70.1%]; absolute difference, 13.1%; 95% CI, 1.0%-24.6%; *P* = .04). However, in the diffuse parenchymal lung disease group, there was no significant difference in diagnostic yield with use of a cryoprobe vs forceps (18 of 25 [72.0%] vs 15 of 24 [62.5%]; absolute difference, 9.5%; 95% CI, −16.0% to 33.6%; *P* = .55) (Table 2; eFigure in Supplement 2). A test result for heterogeneity of effect within diagnostic subgroups was not significant. Final histopathologic diagnosis, localization modalities used during bronchoscopy, and site recruitment data are presented in eTables 1 to 3 in Supplement 2.

Safety

The prespecified combined safety outcome of pneumothorax (grade 2, requiring drainage), bleeding (grade ≥3, requiring balloon tamponade or cessation of the procedure), or respiratory failure occurred in no patients in the cryoprobe group vs 1.6% (4 of 250) in the forceps group (**Table 3**). A total of 6 pneumothoraces were observed, all in the forceps group, 4 of which required chest tube placement; 2 patients were discharged after observation. There were no significant bleeding events (grade ≥3), respiratory failure events, or deaths in the cryoprobe or forceps groups.

The secondary histologic outcomes of biopsy size (total area and total alveolated area) and quality (proportion of total area with crush artifact, proportion of total area with artifact-free lung parenchyma, and histologic accessibility score) are reported in Table 2. Biopsy with a cryoprobe compared with

Table 2. Primary Outcomes and Secondary Analyses

	Cryoprobe (n = 245)	Forceps (n = 245)	Absolute difference (95% CI), %	P value
Primary outcome[a]				
Overall diagnostic yield, No./total No. (%)	217/245 (88.6)	193/245 (78.8)	9.8 (3.3 to 16.3)	.003
Secondary analysis[a]				
Subgroup diagnostic yield, No./total No. (%)				
Lung transplant[b]	120/125 (96.0)	110/124 (88.7)	7.3 (0.6 to 14.4)	.03
Lung nodule or mass[b]	79/95 (83.2)	68/97 (70.1)	13.1 (1.0 to 24.6)	.04
Diffuse lung disease[b]	18/25 (72.0)	15/24 (62.5)	9.5 (−16.0 to 33.6)	.55
Biopsy sample total area, median (IQR), mm^2	16.7 (9.0 to 27.8)	9.6 (5.8 to 16.4)	NA	<.001
Biopsy sample total alveolated area, median (IQR), mm^2	13.1 (5.4 to 23.4)	6.4 (2.9 to 10.9)	NA	<.001
Biopsy sample crush artifact, % of total area, median (IQR)	0 (0 to 5.0)	5 (0 to 10)	NA	<.001
Biopsy sample artifact-free lung parenchyma, % of total area, median (IQR)	100 (95 to 100)	95 (85 to 100)	NA	<.001
Biopsy sample histologic accessibility score, median (IQR)[c]	6 (6 to 6)	6 (5 to 6)	NA	<.001

Abbreviation: NA, not applicable.

[a] All of the samples from each pass were pooled and processed together and were considered a single specimen; each pass was not assessed individually. If there was any diagnostic material in the pooled specimen, the procedure was considered diagnostic.

[b] Post hoc test for heterogeneity of effect within diagnostic subgroups was not significant.

[c] Histologic accessibility index of the biopsy specimen is a 7-point Likert scale of specimen quality; higher scores indicate greater quality.

Table 3. Safety Outcomes

Characteristic, No. (%)	Cryoprobe (n = 250)	Forceps (n = 250)
Combined pneumothorax (grade ≥2), bleeding (grade 3-4), and respiratory failure	0	4 (1.6)
Pneumothorax grade[a]		
1	0	6 (2.4)
2	0	4 (1.6)
Bleeding grade[b]		
1	64 (25.6)	54 (21.6)
2	17 (6.8)	19 (7.6)
3-4	0	0
Respiratory failure	0	0

[a] Pneumothorax grade 1: asymptomatic, clinical or diagnostic observation only; pneumothorax grade 2: symptomatic, intervention indicated (eg, chest tube placement).

[b] Bleeding grade 1: requiring suction to clear; bleeding grade 2: requiring wedging, iced saline, or both; bleeding grade 3: requiring bronchial blocker; bleeding grade 4: cardiopulmonary instability.

forceps yielded significantly larger specimen size (mean [SD], 24.2 [36.7] vs 13.1 [16.6] mm^2; $P < .001$), a higher histologic accessibility score (mean [SD], 5.6 [1.2] vs 5.2 [1.6]; $P < .001$), and less crush artifact (mean [SD], 2.7% [5.3%] vs 8.9% [11.9%]; $P < .001$).

Exploratory Analysis

A post hoc sensitivity analysis performed to account for missing data (eTables 4-7 in Supplement 2) did not alter the primary outcome of overall diagnostic yield. For the lung nodule or mass group, imputations presuming all missing data in the cryoprobe group were nondiagnostic resulted in loss of significance.

Discussion

This is the first randomized clinical trial, to our knowledge, comparing the diagnostic yield of transbronchial biopsy performed with a 1.1-mm cryoprobe vs conventional 2.0-mm forceps. Building on prior evidence of the feasibility[11] and safety[13] of transbronchial biopsy with use of a cryoprobe, the results demonstrate that the diagnostic yield of transbronchial biopsy with cryoprobe was significantly higher than that with forceps in a group of patients with lung nodule or mass, lung transplant, and diffuse parenchymal lung disease. However, among the key secondary outcomes, the diagnostic yield was significantly higher with a cryoprobe vs forceps in the subgroups of nodule or mass and lung transplant, but not in the group with diffuse parenchymal lung disease.

The increased diagnostic yield of the 1.1-mm cryoprobe likely occurred because, compared with forceps, lung tissue specimens obtained with the cryoprobe are larger and have less crush artifact, enhancing cellular yield. Prior comparative studies of the 1.1-mm cryoprobe, although mostly smaller, retrospective, and observational, demonstrated improved diagnostic yield compared with forceps,[32-34] which aligns with the current trial findings.

Additionally, the 1.1-mm cryoprobe was specifically designed to address safety concerns with the required en bloc removal of the bronchoscope and specimen with the larger 1.9- and 2.4-mm cryoprobes.[10] Similar to the FROSTBITE-1 trial,[13] the FROSTBITE-2 trial demonstrated good safety outcomes, with no reported pneumothorax, significant bleeding events, respiratory failure events, or deaths with use of the 1.1-mm cryoprobe.

The FROSTBITE-2 trial has several strengths. First, it included multicenter enrollment of consecutive patients with lung nodule or mass, diffuse parenchymal lung disease, or post-lung transplant referral for transbronchial biopsy. This trial incorporated all populations for which a transbronchial biopsy is indicated, decreasing referral bias and increasing external validity. Second, the primary outcome was determined by an independent, masked pathology core laboratory using strict definitions for a diagnostic yield in each of the 3 subgroups, which reduced site-specific variance in local pathology interpretation and potential for observer bias. Third, the trial was designed to align with routine clinical workflow, integrating eligibility assessment, consent, and study procedures into standard care processes to minimize variation in allocation and reduce loss to follow-up.

Limitations

This study has some limitations. First, the trial was conducted at large US academic centers and the procedure was primarily performed by expert bronchoscopists, so results may not be generalizable to all settings and bronchoscopists. Second, given the nature of the intervention, the bronchoscopists and treating team were aware of the treatment assignments. However, local pathologists and core laboratory, which assessed the primary outcome, were masked to study group assignments. Third, use of a centralized pathologic core for histopathologic assessment may reduce external validity because access to expert pulmonary pathologists may be limited. Fourth, the disposable cryobiopsy equipment is more expensive than forceps, and cost-effectiveness was not addressed in this study.

Conclusions

Transbronchial lung biopsy performed with a 1.1-mm cryoprobe had a significantly higher diagnostic yield compared with 2.0-mm forceps in a group of patients with lung nodules or masses, lung transplant, and diffuse parenchymal lung disease.

ARTICLE INFORMATION

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